Shock modeling with UCLCHEM

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A short introduction

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Parametrization of C-shocks. Evolution of the sputtering of grains

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Tracing shock type with chemical diagnostics

An application to L1157

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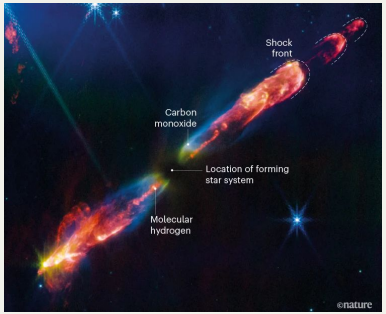
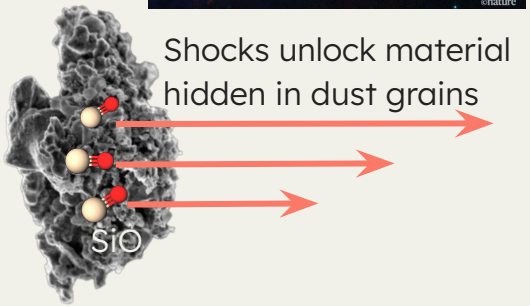
Shock chemistry: why should we care?

Shocks reshape chemistry on short timescales

- ➔ Rapid heating (10 → 1000+ K)
- ➔ Gas compression (↑ density)

- ➔ **Grain processing** (sputtering, mantle removal)
  - ◆ Inject new species into the gas phase
- ➔ Activate reactions that are inefficient in cold gas
  - ◆ High-temperature chemistry becomes efficient





Shocks unlock material hidden in dust grains

How UCLCHEM models shocks

UCLCHEM models the chemistry in two stages

Pre-shock evolution

- ➔ Cold, dense gas (formation of the cloud)
- ➔ Chemistry evolves over time
- ➔ Builds up ice mantles on grains

=
initial conditions

time →

Shock

- ➔ Temperature ↑↓
- ➔ Density ↑
- ➔ Grain processing (sputtering)
- ➔ Chemistry responds to shock conditions

=
shock + post-shock evolution

Parameterized physical evolution + time-dependent chemistry

What do we control in UCLCHEM?

Shock models are defined by:
shock type + a small set of parameters

Step 1: Shock type

C-type
(continuous)

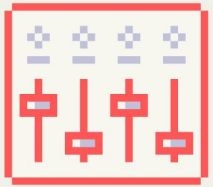
$t \leq t_{diss}$	$t > t_{diss}$
shock	post-shock

J-type
(jump)

$t = 0$	$t = t_{shock}$	$t = t_{cool}$
shock front	cooling phase	post-shock gas

then choose parameters

Step 2: Parameters

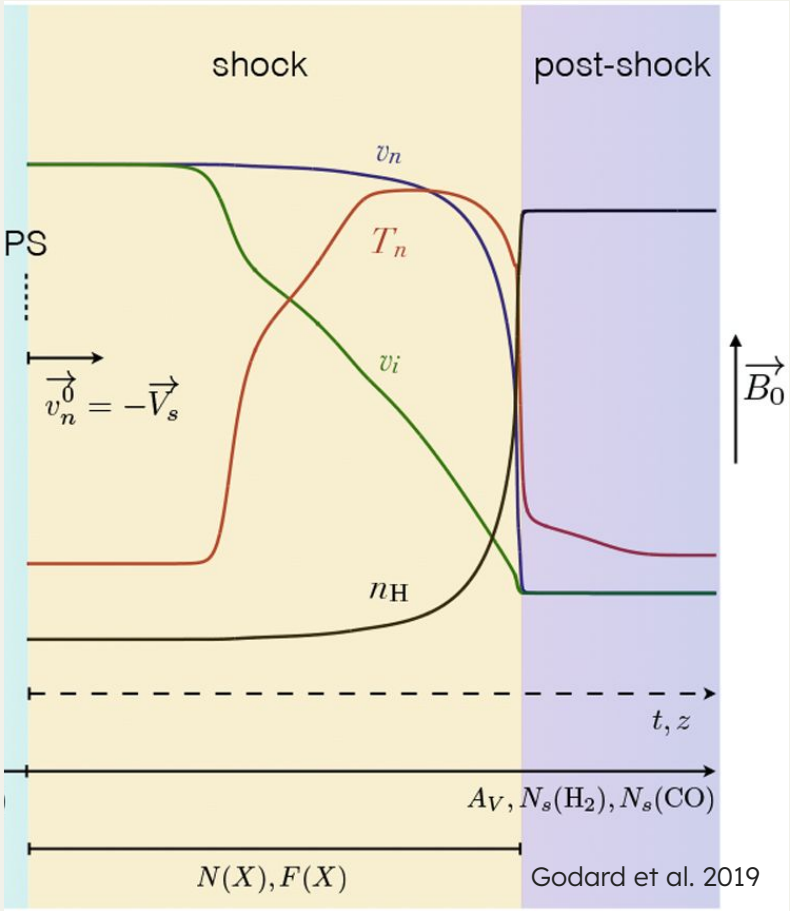


shock velocity
pre-shock density

- ➔ magnetic field (`bm0`, for the C-type)
- ➔ other parameters (fixed for this tutorial)

$v_s \uparrow \rightarrow T \uparrow \rightarrow$ sputtering $\uparrow \rightarrow$ SiO $\uparrow$

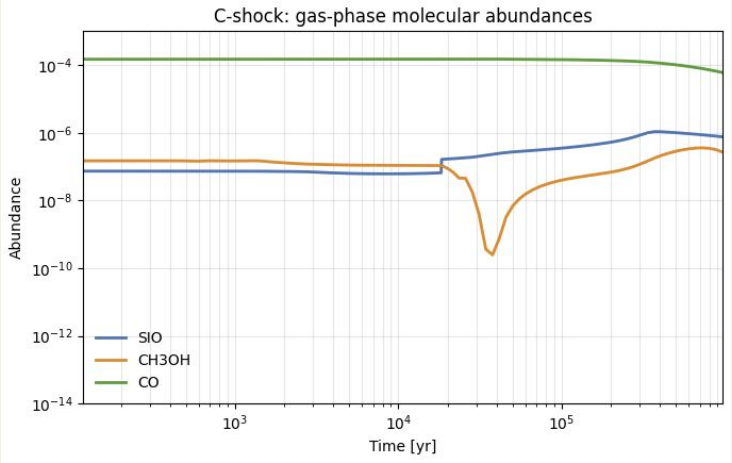
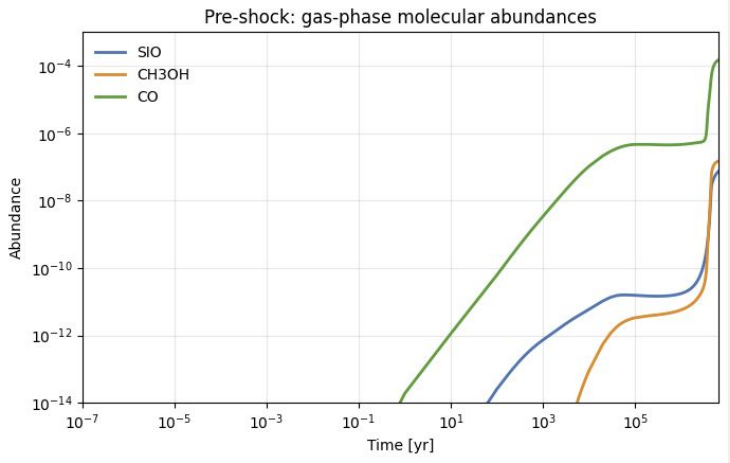
Shock and post-shock



Godard et al. 2019

```
cshock = uclchem.model.CShock(shock_vel=40, param_dict=param_dict)
dissipation_time = cshock.dissipation_time
```

What will we do in the tutorial?

We will:

1. Run a pre-shock model
2. Compare C-type and J-type shocks
3. Explore the effect of density
4. Explore the effect of shock velocity